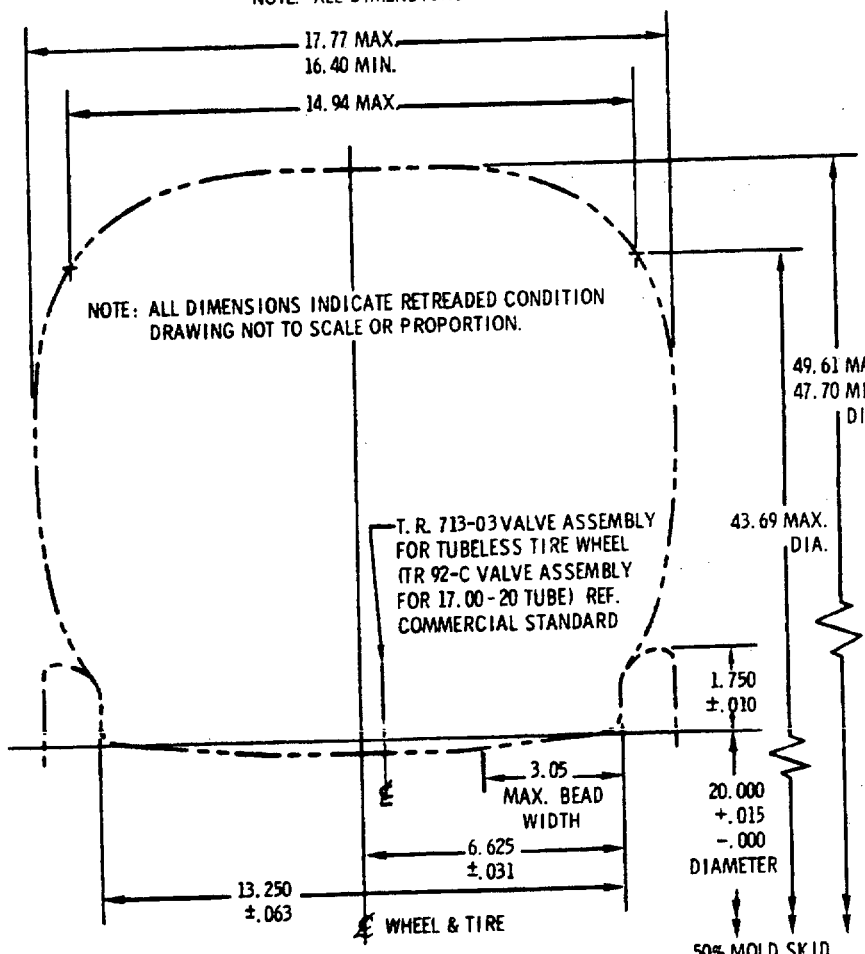
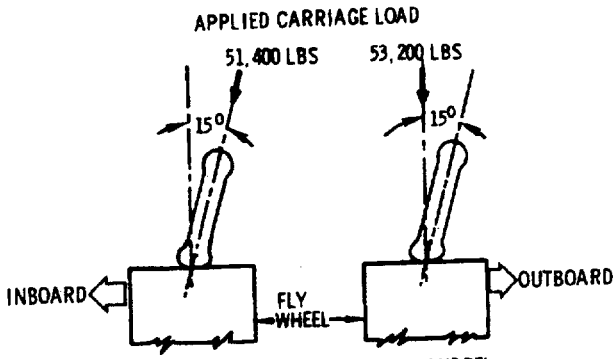
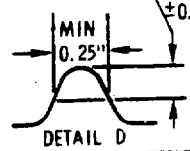
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NOTE: ALL DIMENSIONS ARE IN INCHES







DETAIL D

FLYWHEEL DIAMETER (IN)	26 PR TEST INFL PRESSURE (PSI)
120	197
96	210
84	216

TEST INFLATION PRESSURE
DETAIL C

FIGURE 1 Ⓐ FOR CHANGES SEE SHEET 1, 2, & 3

APPROVED 21 DEC. 1970 REVISED Ⓐ 1 APRIL 1978

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P.A. 99 Other Cost	TITLE: TIRE - PNEUMATIC RETREAD, TYPE VII 49 X 17 / 26 PR	MILITARY STANDARD MS27811 (USAF)
PROCUREMENT SPECIFICATION: MIL - R - 7726	SUPERSEDES: AF DWG NO. 62D3107	SHEET 1 OF 3

DD FORM 1 SEP 65 672-1 (Limited coordination)

EDITION OF 1 SEP 67 MAY BE USED.

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GENERAL NOTES

1. ANY DEVIATION FROM GIVEN DIMENSIONS MUST BE AUTHORIZED BY WRITTEN APPROVAL OF THE PROCURING AGENCY. DIMENSIONS CALLED OUT ARE AFTER RETREADING.

2. REFERENCE TO ANY SPECIFICATION HEREIN SHOULD BE CONSTRUED AS REFERENCE TO THE ISSUE IN EFFECT ON THE DATE OF INVITATIONS FOR BIDS.

3. WHERE CONFLICT EXISTS BETWEEN THIS STANDARD AND ANY OTHER MILITARY SPECIFICATION, THE REQUIREMENTS OF THIS STANDARD SHALL APPLY.

4. THE QUALITY ASSURANCE SHALL BE IN ACCORDANCE WITH REQUIREMENTS OF MIL-R-7726.

RETREADED TIRE NOTES

THE RETREADED TIRE SHALL CONFORM TO THE APPLICABLE REQUIREMENTS OF MIL-R-7726 AND SHALL COMPLY WITH THE REQUIREMENTS LISTED BELOW:

SIZE	PLY RATING	MAX STATIC LOAD (LBS)	INFLATION PRESSURE PSI (RATED)	BEAD WIDTH (MAX) IN *	WEIGHT LB (MAX)	STATIC UNBALANCE IN OZ (MAX)	TREAD	MOLD SKID DEPTH (MIN) IN
49x17	26	39,600	170	8.05	226	40	RIB	0.40

*EXCLUDING MAXIMUM BEAD TOE FLASH OF 3/8 INCHES

1. THREE (3) CONSECUTIVE TIRES ARE TO SUCCESSFULLY COMPLETE THE DYNAMOMETER TESTING. THE FIRST TIRE SHALL BE TESTED TO THE ENTIRE TEST SPECTRUM UNDER TIRE NOTE 2: THE REMAINING TWO TIRES ARE TO BE TESTED TO TEST CONDITION "A_1" (HIGH SPEED TAXI-TAKE-OFF) AND "A_2", (LOW SPEED TAXI-TAKE-OFF).

2. THE TIRE SHALL SATISFACTORILY WITHSTAND 20 CYCLES OF TEST "A_1" 50 CYCLES OF TEST "A_2", 70 CYCLES OF TEST "B", 10 CYCLES OF TEST "C", 60 CYCLES OF TEST "D" (30 CYCLES D_1, 30 CYCLES D_2). TESTS "C" AND "D" SHALL BE PERFORMED LAST WITH TEST "C" PRECEDING "D". THE SEQUENCE OF TESTING FOR "A" AND "B" IS OPTIONAL.

A_1 HIGH SPEED TAXI-TAKE-OFF

THE TIRE SHALL BE LANDED AGAINST A STATIONARY FLYWHEEL AT THE MAXIMUM STATIC LOAD, THE FLYWHEEL SHALL THEN BE ACCELERATED TO 30 MPH AND MAINTAINED AT THIS VELOCITY UNTIL A TAXI ROLL DISTANCE OF 13,500 FEET HAS BEEN COVERED. THE FLYWHEEL SHALL THEN BE DECELERATED TO ZERO MPH AND IMMEDIATELY ACCELERATED AT 4FT/SEC/SEC AVERAGE (SIMULATING TAKE-OFF) TO 225 MPH. DURING THE ACCELERATION THE LOAD SHALL BE MAINTAINED AT THE MAXIMUM STATIC LOAD FOR THE FIRST 20 SEC AND THEN DECREASED LINEARLY WITH TIME TO 22,000 LBS IN 75 SEC AFTER ACCELERATION STARTS, THEN REDUCED TO ZERO LOAD AFTER A ROLL DISTANCE OF 13,500 ± 200 FT HAS BEEN COVERED IN APPROXIMATELY 82.5 SECONDS.

A_2 LOW SPEED TAXI-TAKE-OFF

THE TIRE SHALL BE LANDED AGAINST A STATIONARY FLYWHEEL AT 34,500 LBS. THE FLYWHEEL SHALL THEN BE ACCELERATED TO 30 MPH AND MAINTAINED AT THIS VELOCITY UNTIL A TAXI ROLL DISTANCE OF 13,500 FEET HAS BEEN COVERED. THE FLYWHEEL SHALL THEN BE DECELERATED TO ZERO MPH AND IMMEDIATELY ACCELERATED AT 4.30 FT/SEC/SEC (SIMULATING TAKEOFF) TO 214 MPH. DURING THE ACCELERATION THE LOAD SHALL BE MAINTAINED AT THE 34,500 LBS, FOR THE FIRST 20 SECONDS AND THEN DECREASED LINEARLY WITH TIME TO 19,500 LBS IN 66 SECONDS AFTER ACCELERATION STARTS, THEN REDUCED TO ZERO LOAD AFTER A ROLL DISTANCE OF 11,500 FT ± 200 FT HAS BEEN COVERED IN APPROXIMATELY 74 SECONDS.

B. LANDING - TAXI

THE TIRE SHALL BE LANDED AGAINST A FLYWHEEL ROTATING AT A PERIPHERAL SPEED OF 160 MPH. THE FLYWHEEL SHALL BE DECELERATED AT AN AVERAGE RATE OF 4.5 FT/SEC/SEC BETWEEN 160 AND 15 MPH IN A ROLL DISTANCE OF 6,100 ± 200 FT. AN AIRCRAFT BRAKE MAY BE USED TO OBTAIN THE REQUIRED DECELERATION. THE LOAD SHALL BE INCREASED TO 15,000 LBS 8 SECONDS AFTER LANDING THEN FURTHER INCREASED TO 23,500 LBS APPROXIMATELY 47.3 SECONDS AFTER LANDING. THE FLYWHEEL SHALL THEN BE ACCELERATED TO 30 MPH AND THE LOAD SHALL REMAIN CONSTANT UNTIL A TAXI ROLL DISTANCE OF 13,500 ± 200 FT HAS BEEN COVERED.

C. TAXI - ROLL

THE FLYWHEEL SHALL BE ACCELERATED TO 30 MPH AND TAXIED FOR 35,000 FT AT THE MAXIMUM STATIC LOAD. THE TIRE SHALL BE ALLOWED TO COOL TO 120° ± 10°F BEFORE REPEATING THE CYCLE.

D. TAXI - TURN (NOT APPLICABLE TO 24 PR TIRES)

CONDUCT TAXI-TURN IN ACCORDANCE WITH THE "MOVEABLE CARRIAGE" PROCEDURE (FIGURE 1, DETAIL A). IF THIS CAPABILITY DOES NOT EXIST AN ACCEPTABLE ALTERNATE IS THE "BENT MANDREL" PROCEDURE (DETAIL B).

APPROVED 21 DEC 1970 REVISED Ⓐ FOR CHANGES SEE SHEET 1, 2, & 3

P.A. 99 Other Cust	TITLE TIRE - PNEUMATIC RETREAD, TYPE VII 49 X 17 / 26 PR	MILITARY STANDARD MS27811 (USAF)
PROCUREMENT SPECIFICATION MIL-R-7726	SUPERSEDES AF DWG NO. 62D3107	SHEET 2 OF 3

DD FORM 672-1 (Limited coordination)

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D_1 TAXI-TURN-OUTBOARD

1. ACCELERATE THE FLYWHEEL TO 15 MPH AND MAINTAIN THIS SPEED THROUGHOUT THE ENTIRE CYCLE.

2. LAND THE TIRE ON THE FLYWHEEL IN A POSITION OF 15° CAMBER OUTBOARD.

3. INCREASE THE CARRIAGE LOAD LINEARLY WITH TIME FROM ZERO TO 51,400 LBS FOR DETAIL A OR 53,200 LBS FOR DETAIL B, WITHIN 2 TO 3 SECONDS AFTER LANDING AND MAINTAIN THIS LOAD FOR REMAINDER OF CYCLE.

4. UNLAND THE TIRE WHEN A DISTANCE OF 2,300 FT HAS BEEN COVERED.

D_2 TAXI-TURN-INBOARD

SAME AS D_1 EXCEPT LAND THE TIRE IN A POSITION OF 15° CAMBER INBOARD.

3. TEST INFLATION PRESSURE. THE TIRE INFLATION PRESSURE USED FOR ALL CYCLES SHALL BE IN ACCORDANCE WITH DETAIL C FOR THE RESPECTIVE DIAMETER USED TO CONDUCT THE CYCLE(S). THE TIRE OUTSIDE DIAMETER, ROLLING RADIUS AND DEFLECTION ON THE FLYWHEEL AT RATED LOAD AND TEST INFLATION PRESSURE FOR THE CORRESPONDING FLYWHEEL DIAMETER SHALL BE RECORDED AND STATED IN THE QUALIFICATION TEST REPORT AFTER A MINIMUM OF 12 HOURS OF STATIC GROWTH AT THE RATED INFLATION PRESSURE.

4. THE RETREADED TIRE SHALL FUNCTION INTERCHANGEABLY AS EITHER A TUBE OR TUBELESS TYPE TIRE. THE TIRE MAY BE CONSTRUCTED WITH A FABRIC REINFORCED TREAD DESIGN FOR HIGH SPEED OPERATION AND MAXIMUM RELIABILITY AND RETENTION OF THE TREAD TO THE CARCASS.

5. THE TREAD PATTERN SHALL HAVE A MINIMUM OF FOUR AND A MAXIMUM OF SEVEN GROOVES. THESE GROOVES SHALL BE CONTINUOUS AND CIRCUMFERENTIAL. IF THE TREAD PATTERN HAS FIVE OR MORE GROOVES, THE TWO OUTER GROOVES MAY BE OF PARTIAL SKID DEPTH (MINIMUM OF 60% OF THE ACTUAL MOLD SKID DEPTH). ALL OTHER GROOVES SHALL BE A MINIMUM OF THE SKID DEPTH SPECIFIED ABOVE. A MINIMUM OF FOUR GROOVES SHALL BE WITHIN THE CONTACT AREA OF THE STATIC LOAD FOOTPRINT TAKEN AT 60% OF THE MAXIMUM STATIC LOAD AT RATED INFLATION PRESSURE. THE MOLD GROOVE WIDTH SHALL BE A MINIMUM OF .25 INCHES MEASURED AS SHOWN IN DETAIL D. THE WIDTH SHALL BE REPORTED IN THE QUALIFICATION TEST REPORT. THE MAXIMUM CENTER RIB WIDTH OF THE UNINFLATED TIRE SHALL NOT EXCEED 3.0 INCHES.

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P.A. 99 Other Cont	TITLE TIRE - PNEUMATIC RETREAD, TYPE VII 49 X 17/26 PR	MILITARY STANDARD MS 27811 (USAF)
PROCUREMENT SPECIFICATION MIL-R-7726	SUPERSEDES A.F DWG NO. 62D3107	SHEET 3 OF 3

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☆U.S. GOVERNMENT PRINTING OFFICE 1978—703-122/2270

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